

Statistical Design and Models for Evaluation Studies (37495)

Lecture 4 - Comparing Treatments

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1. Lecture outline

Topics:

- Contrasts
- Multiple contrast testing
 - ① Fisher's LSD method
 - ② Scheffe's method
 - ③ Tukey's HSD test
 - ④ Dunnett's test
- Example 1 in R
- Example 2 in R

- ANOVA F test result: gives a “Yes/No” answer to the question “is there any treatment effect at all?”
- Often we have more specific question e.g. compare a set of new treatments vs a control treatment or do pairwise comparisons between many (or all) treatments
- Next step: identify and classify treatments into groups of treatments that are not significantly different
- Formulate such questions in terms of contrasts

Contrasts

Suppose we want to compare the effects of treatments corresponding to μ_1 versus that of μ_2 . In the simplest case, we could set up the hypotheses

$$\begin{aligned} H_0 : \mu_1 - \mu_2 &= 0 & H_0 : \mu_1 &= \mu_2 \\ H_1 : \mu_1 - \mu_2 &\neq 0. & H_1 : \mu_1 &\neq \mu_2 \end{aligned}$$

We can encode this with a vector $\lambda = (\lambda_i)_{i=1}^a \in \mathbb{R}^a$ and re-write the hypotheses as

$$\begin{aligned} H_0 : \sum_{i=1}^a \lambda_i \mu_i &= 0 \\ H_1 : \sum_{i=1}^a \lambda_i \mu_i &\neq 0. \end{aligned}$$

a = no. of groups
 $1 \times \mu_1 + (-1) \times \mu_2 = 0$

Here $\lambda_1 = 1$ and $\lambda_2 = -1$.

Contrasts

A more sophisticated example would be $\lambda = (1, -1/2, -1/2, 0, \dots)$ which compares the first treatment to the average of the second and third via the hypotheses

$$H_0 : \mu_1 - (\mu_2 + \mu_3)/2 = 0$$

$$H_1 : \mu_1 - (\mu_2 + \mu_3)/2 \neq 0.$$

$$\mu_1 \approx \frac{\mu_2 + \mu_3}{2}$$

Of course, we could also perform hypothesis tests with non-zero right-hand-sides.

Bottom line comparison of treatments is about testing whether a certain linear combination of treatment means is zero.

Contrasts

A **contrast** is a set of weights $\lambda = (\lambda_i)_{i=1}^a$ of the treatment effects (or means) which sum up to zero, i.e. $\sum_{i=1}^a \lambda_i = 0$.

Given contrast $\lambda = (\lambda_i)_{i=1}^a$, consider the hypothesis $H_0: \sum_i \lambda_i \mu_i = 0$ and define the statistic

$$t_{obs}(\lambda) = \frac{\sum_{i=1}^a \lambda_i \bar{y}_{i.}}{\sqrt{\sum_{i=1}^a \frac{\lambda_i^2}{n_i} MS_{Er}}},$$

*sample mean of
ith group*

where n_i is the number of observations for the i th treatment group (for instance n for equal-sized groups) and recalling $N = \sum_{i=1}^a n_i$ as the total number of observations).

If H_0 holds, then the statistic t_{obs} follows a t -distribution with $N - a$ degrees of freedom.

The square of this statistic

$$f_{obs} := t_{obs}^2(\lambda) = \frac{(\sum_{i=1}^a \lambda_i \bar{y}_{i.})^2}{\sum_{i=1}^a \frac{\lambda_i^2}{n_i} MS_{Er}}$$

follows an F -distribution with $(1, N - a)$ degrees of freedom.

Note that knowing the distribution, any confidence interval for the statistics t_{obs} , f_{obs} can be calculated.

Contrasts - an example

Consider the simple hypotheses

$$H_0 : \mu_i - \mu_j = 0$$

$$H_1 : \mu_i - \mu_j \neq 0. \quad i \neq j$$

$$\lambda_1 = 1$$
$$\lambda_2 = -1$$

In this case, the test statistic is

$$t_{obs}(\lambda) = \frac{\bar{y}_i. - \bar{y}_j.}{\sqrt{MS_{Er} \left(\frac{1}{n_i} + \frac{1}{n_j} \right)}},$$

and a $100 \times (1 - \alpha)$ % confidence interval for $\mu_i - \mu_j$ is

$$\left[(\bar{y}_i. - \bar{y}_j.) - \underline{t_{N-a;1-\alpha/2}} \sqrt{MS_{Er} \left(\frac{1}{n_i} + \frac{1}{n_j} \right)}, \right. \\ \left. (\bar{y}_i. - \bar{y}_j.) + \underline{t_{N-a;1-\alpha/2}} \sqrt{MS_{Er} \left(\frac{1}{n_i} + \frac{1}{n_j} \right)} \right]$$

for any pair of groups (i, j)

Contrasts - an example

Finding the value 0 in such an interval is an indication that μ_i and μ_j are not “significantly” different. On the other hand, if the interval does not contain 0, that is evidence that $\mu_i \neq \mu_j$. However, these confidence intervals are **sometimes**

misleading and must be **carefully analysed**, in particular when related to the global null hypothesis $H_0 : \mu_1 = \mu_2 = \dots = \mu_a$



$a > 2$

Remark

Two contrasts with weights $\lambda = (\lambda_i)_{i=1}^a$ and $\nu = (\nu_i)_{i=1}^a$ are said to be *orthogonal* if $\sum_{i=1}^a \frac{\lambda_i \nu_i}{n_i} = 0$ holds. It turns out that statistics t_{obs} corresponding to orthogonal contrasts are independent.

This observation is important in the procedures of subsequent testing of multiple hypothesis corresponding to different contrasts.

Multiple contrast testing

Problem with all statistical tests: (overall) error rate increases with the increasing number of tests.

We know the probability of a type I error for a single test is α . Suppose that

$H_0: \mu_1 = \mu_2 = \dots = \mu_a$ is true.

$P(\text{reject } H_0 | H_0 \text{ is true})$

Assume we perform K pairwise tests $H_0^{(i,j)}: \mu_i = \mu_j$ using an individual significance level of α and the tests were *independent* (which they are not), we would **wrongly reject** at least one null hypothesis with probability $1 - (1 - \alpha)^K$

$$1 - P(\text{all right decisions}) = 1 - P(\text{not reject } H_0 | H_0 \text{ is true}) = 1 - (1 - \alpha)^K$$

If the decisions were *perfectly dependent* (which they are not either), we would **wrongly reject** at least one null hypothesis with probability α

$\Rightarrow$ if we based our decision about $H_0: \mu_1 = \mu_2 = \dots = \mu_a$ on the pairwise comparison tests, we would **wrongly reject** H_0 with a probability strictly between α and $1 - (1 - \alpha)^K$, larger than α

Hence, we need a way to control the overall error rate.

Multiple contrast testing

In the setting of ANOVA, we try to compare different treatments and conclude there is a significant difference if the corresponding hypothesis about a contrast can be rejected.

Suppose $a = 4$ groups, which would give ~~k~~ $= \binom{4}{2} = 6$ pairwise comparisons. The test based on pairwise comparisons would be of effective significance level between 0.05 and $1 - (1 - 0.05)^6 = 0.265$, larger than 0.05.

It is usually not possible to determine exactly the significance level of such a test: it all depends on the exact level of dependence between the decisions about the different pairwise comparisons.

⇒ Need methodologies to allow simultaneous comparison of different treatments with a built in correction, even for a large family of contrasts being tested

Multiple contrast testing

Define **family-wise error rate** as the probability of rejecting at least one of the hypotheses under the assumption that all are true. We say that a procedure controls the family-wise error rate in the at level α if the family-wise error rate is $\leq$ to significance level α , with a typical choice $\alpha = 0.05$.

We will now look at some treatment comparison methods with a built-in control of family-wise error rate.

Some methods also return simultaneous confidence intervals at level $1 - \alpha$ and we reject the null hypothesis with controlled family-wise error rate α .

Fishers' Least Significance Difference (LSD) Method

- Method which does not control for family-wise error rate, i.e. the significance level α is individual, per comparison
- So even there is no significant difference in treatments, the probability that at least one pair will be declared to have a significant difference is very high, even for small values of α
- Formula for the LSD: $LSD_{i,j} = t_{0.025, df_{Er}} \sqrt{MS_{Er}(1/n_i) + (1/n_j)}$
- If $|\bar{y}_i - \bar{y}_j| \geq LSD_{i,j}$ then we reject the null hypothesis $H_0 : \mu_i = \mu_j$

Scheffe's test

- Controls the family-wise error rate for all possible contrasts
- Has low statistical power
- The Scheffé procedure works as follows: calculate F -ratio use quantiles of the distribution $F_{a-1, N-a}$ to calculate p-values or critical regions. More precisely, for each contrast $(\lambda_i)_{i=1}^a$ the hypothesis

$$H_0 : \sum_{i=1}^a \lambda_i \mu_i = 0$$

can be rejected at significance level α if

$$\frac{(\sum_i \lambda_i \bar{y}_{i.})^2}{(a-1)MSE \sum_i \lambda_i^2 / n_i} \geq f_{a-1, N-a, 1-\alpha}$$

- For a simple comparison $H_0 : \mu_i - \mu_j = 0$, this means we reject H_0 if

$$|\bar{y}_{i.} - \bar{y}_{j.}| \geq \sqrt{(a-1)f_{a-1, N-a, 1-\alpha} MS_{Er}(1/n_i + 1/n_j)}$$

Tukey's Honestly Significant Difference (HSD) test

- Based on **studentised range distribution**
- Controls for family-wise error rate for all pairwise contrasts with hypotheses of the form $H_0 : \mu_i - \mu_j = 0, i \neq j$
- If the treatment groups have the same size i.e. $n_i = n$ for all $i = 1, \dots, a$, then the simultaneous confidence intervals hold with probability $1 - \alpha$ (if unequal sample sizes then it is at least this)
- Studentised range is defined by

$$\frac{\bar{y}_{max} - \bar{y}_{min}}{\sqrt{MS_{Er}/n}},$$

where $\bar{y}_{max}$ and $\bar{y}_{min}$ are the smallest and largest sample means.

- The distribution of the studentised range under H_0 (all treatments have same effect) follows a distribution $Q_{a,N-a}$ which depends on the number of groups a and on the degrees of freedom $N - a$ of MSE . Denote its $1 - \alpha$ -quantile by $q_{a,N-a,1-\alpha}$
- Tukey's test: two treatment means are significantly different if

$$|\bar{y}_i - \bar{y}_j| \geq q_{a,N-a,1-\alpha} \sqrt{MS_{Er}/n}$$

- Tukey-Kramer method allows for unequal sample sizes between treatments

Dunnett's test

$$\alpha = 4 \quad \binom{4}{2} = 6$$

control | A B C

- Used to compare all of the treatments to a single control treatment (“many-to-one comparisons”)
- Modification of the usual t -test by using a different critical value
- Family-wise error rate for those tests is α

Example 1

The data below are reported in Cochran and Cox (1957) and show the results of an experiment on the effects of applications of sulphur in reducing scab disease in potatoes. The objective in applying sulphur is to increase the acidity of the soil since scab does not thrive in very acidic soil.

In addition to untreated plots, which serve as a control, three amounts of dressing were applied: 300, 600 and 1200 pounds per acre. Both a fall and a spring application of each amount was tested, so that in all there were 7 distinct treatments. The sulphur was spread by hand on the surface of the soil and then disced to a depth of about 4 inches.

The quantity to be analysed is the “scab index”. This is, roughly speaking, the percentage of the surface area of the potato that is infected with scab. It is obtained by examining 100 potatoes at random from each plot, grading each potato on a scale from 0 to 100% infected and taking the average.

Example 1

Group		Observations							
Control		12	10	24	29	30	18	32	26
Spring 300		30	7	21	9				
Spring 600		18	24	12	19				
Spring 1200		17	7	16	17				
Autumn 300		9	9	16	4				
Autumn 600		16	10	18	18				
Autumn 1200		10	4	4	5				

Potato scab data

Example 1 in R

```
# Response and treatment data
data1 <- c("Control",12,"Control",10,"Control",24,"Control",29,
"Control",30, "Control",18,"Control",32,"Control",26,"S300",30,
"S300",7,"S300",21,"S300",9,"S600",18,"S600",24,"S600",12,
"S600",19,"S1200",17,"S1200",7,"S1200",16,"S1200",17,"A300",
9,"A300",9,"A300",16,"A300",4,"A600",16, "A600",10,"A600",
18,"A600",18,"A1200",10,"A1200",4,"A1200",4,"A1200",5)
head(data1) #view first six elements
# [1] "Control" "12"          "Control" "10"          "Control" "24"
```

Note: vectors can only have elements of the same type, so numeric values have been converted to characters. When we save in a data frame, R will automatically convert character mode to factor. This is fine for the treatments, but we will have to convert the response data to numeric.

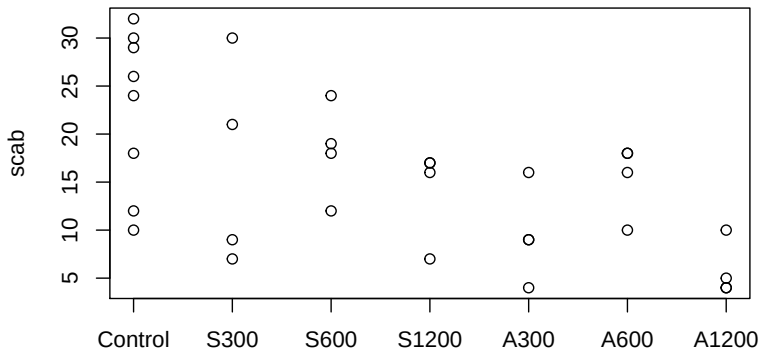
Example 1 in R

```
potato <- data.frame(data1[seq(from=1, to=length(data1)-1, by=2)],  
  as.numeric(data1[seq(from=2, to=length(data1), by=2)]))  
names(potato) <- c("treatment", "scab") #change names of columns  
sapply(potato, class) #check if type-conversion worked  
# treatment      scab  
# "factor" "numeric"  
# Re-order levels and make Control default (alphabetical order is default)  
potato$treatment <- factor(potato$treatment,  
  levels = c("Control", "S300","S600", "S1200", "A300", "A600",  
  "A1200"))
```

Example 1 in R

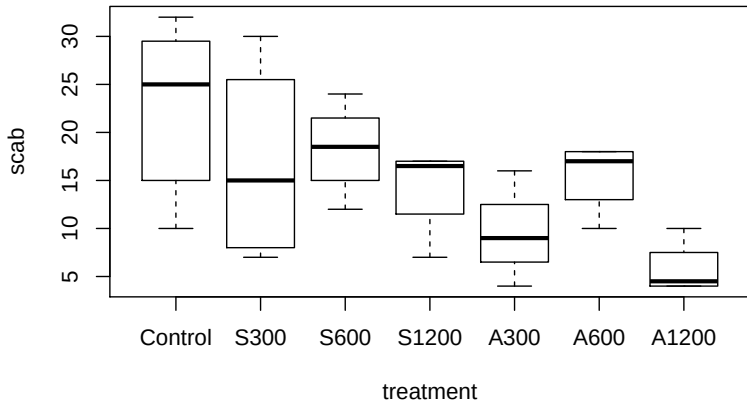
```
# Plot data
```

```
stripchart(scab~treatment, vertical=TRUE, pch=1, data=potato)
```



Example 1 in R

```
boxplot(scab~treatment, data=potato)
```



Example 1 in R

$$H_0: \mu_1 = \mu_2 = \dots = \mu_7$$

Now, for the ANOVA analysis and view diagnostic plots.

```
# Fit ANOVA model
```

```
fit.potato <- aov(scab~treatment, data=potato)
```

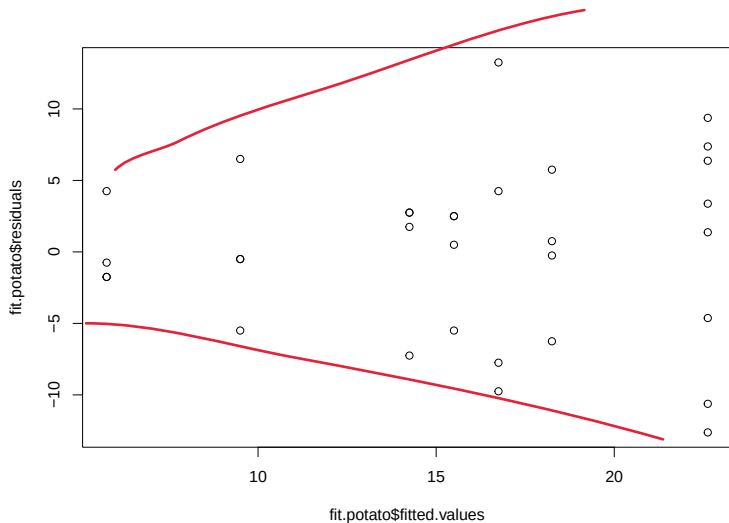
```
summary(fit.potato)
```

#		Df	Sum Sq	Mean Sq	F value	Pr(>F)
# treatment		6	972.3	162.06	3.608	0.0103 *
# Residuals		25	1122.9	44.92		
# ---						
# Signif. codes:		0 '***'	0.001	'**'	0.01	'*' 0.05 '.' 0.1 ' ' 1

Handwritten notes: A red circle is drawn around the p-value 0.0103. To the right of the circle, there is a red arrow pointing to the p-value with the text "< 0.05" and "Reject H₀". Below the circle, the word "Reject" is written in red.

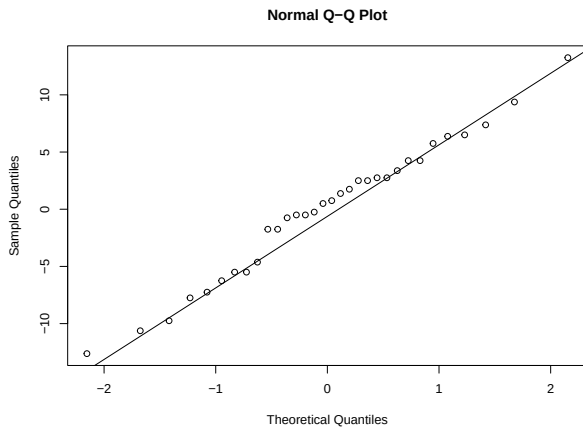
Example 1 in R

```
plot(fit.potato$residuals~fit.potato$fitted.values)
```



Example 1 in R

```
qqnorm(fit.potato$residuals)  
qqline(fit.potato$residuals)
```



Example 1 in R

$$H_0 : \tau_1 = \tau_2 \cdots \tau_7 = 0$$

$$H_1 : \text{at least one } \tau_i \neq 0.$$

The null hypothesis can be rejected with p-value 0.0103 and we conclude at least one treatment effect is non-zero (or equivalently at least one mean is different from the others). However, it does look like a problem with increasing variance. We can test this with Levene's test of equal variances.

Example 1 in R

```
tapply(potato$scab, potato$treatment, var) #view sample variances
#   Control      S300      S600      S1200      A300      A600      A1200
# 69.98214 116.25000  24.25000  23.58333  24.33333  14.33333  8.25000
library("lawstat") # install.packages("lawstat") #install if necessary
levene.test(potato$scab, potato$treatment, location="median",
            correction.method="zero.correction")
# Modified robust Brown-Forsythe Levene-type test based on the absolute
# deviations from the median with modified structural zero removal
method
# and correction factor
#
# data:  potato$scab
# Test Statistic = 2.1743, p-value = 0.09423
```

Sample variances are quite different and Levene's test is marginally insignificant. A Box-Cox transform could assist, but we will proceed with the current model.

What about the normality assumption? We test this next.

Example 1 in R

```
res <- residuals(fit.potato)
# Shapiro-Wilk test
shapiro.test(res)
# Shapiro-Wilk normality test
#
# data:  res
# W = 0.98267, p-value = 0.8724
#
# Kolmogorov-Smirnov test
ks.test(res, "pnorm" , mean=mean(res), sd=sd(res))
# One-sample Kolmogorov-Smirnov test
#
# data:  res
# D = 0.10666, p-value = 0.8598
# alternative hypothesis: two-sided
#
# Warning message:
# In ks.test(res, "pnorm", mean = mean(res), sd = sd(res)) :
# ties should not be present for the Kolmogorov-Smirnov test
```

Example 1 in R

```
# Cramer-von Mises test
library(nortest) # install.packages("nortest") #install if necessary
nortest::cvm.test(res)
# Cramer-von Mises normality test
#
# data:  res
# W = 0.05288, p-value = 0.4581
#
# Anderson-Darling test
nortest::ad.test(res)
# Anderson-Darling normality test
#
# data:  res
# A = 0.28405, p-value = 0.6083
We see there is no problem with the normality assumption.
```

Example 1 in R

We now turn to contrasts. Let's start with Tukey's method

CIs with emmeans

library("emmeans") # install.packages("emmeans") #install if necessary

meansObj <- emmeans(fit.potato, "treatment")

confint(meansObj, level=0.95) #individual CIs

#	treatment	emmean	SE	df	lower.CL	upper.CL
#	Control	22.62	2.37	25	17.74	27.5
#	S300	16.75	3.35	25	9.85	23.7
#	S600	18.25	3.35	25	11.35	25.2
#	S1200	14.25	3.35	25	7.35	21.2
#	A300	9.50	3.35	25	2.60	16.4
#	A600	15.50	3.35	25	8.60	22.4
#	A1200	5.75	3.35	25	-1.15	12.7

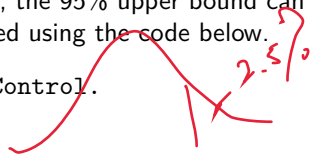
est. $\pm$ quantile $\times$ S.E

The CIs shown above are the individual CIs on the population means associated with the samples. That is, family-wise error rate has not been controlled.

Example 1 in R

The CIs are calculated in the usual way; for instance, the 95% upper bound can for the Control population mean (μ_1) can be obtained using the code below.

```
# Individual 95% upper bound for mean of Control.  
22.62 + 2.37*qt(0.975, 25)  
# [1] 27.50111
```



We can use the `emmeans` object to obtain CIs on contrasts where the family-wise error rate has been controlled. First we look at all pairwise comparisons using Tukey's method.

Example 1 in R

```
# Tukey's pairwise comparisons
fit.tukey <- contrast(object=meansObj, method="pairwise", adjust="tukey")
summary(fit.tukey, infer=c(T,T), level=0.95, side="two-sided") #view results
```

# contrast	estimate	SE	df	lower.CL	upper.CL	t.ratio	p.value
# Control - S300	5.88	4.10	25	-7.25875	19.01	1.432	0.7801
# Control - S600	4.38	4.10	25	-8.75875	17.51	1.066	0.9322
# Control - S1200	8.38	4.10	25	-4.75875	21.51	2.041	0.4158
# Control - A300	13.12	4.10	25	-0.00875	26.26	3.198	0.0502
# Control - A600	7.12	4.10	25	-6.00875	20.26	1.736	0.5998
# Control - A1200	16.88	4.10	25	3.74125	30.01	4.112	0.0060
# S300 - S600	-1.50	4.74	25	-16.66555	13.67	-0.317	0.9999
# S300 - S1200	2.50	4.74	25	-12.66555	17.67	0.528	0.9981
# S300 - A300	7.25	4.74	25	-7.91555	22.42	1.530	0.7250
# S300 - A600	1.25	4.74	25	-13.91555	16.42	0.264	1.0000
# S300 - A1200	11.00	4.74	25	-4.16555	26.17	2.321	0.2733
# S600 - S1200	4.00	4.74	25	-11.16555	19.17	0.844	0.9775
# S600 - A300	8.75	4.74	25	-6.41555	23.92	1.846	0.5314
# S600 - A600	2.75	4.74	25	-12.41555	17.92	0.580	0.9969
# S600 - A1200	12.50	4.74	25	-2.66555	27.67	2.638	0.1571
# S1200 - A300	4.75	4.74	25	-10.41555	19.92	1.002	0.9487
# S1200 - A600	-1.25	4.74	25	-16.41555	13.92	-0.264	1.0000
# S1200 - A1200	8.50	4.74	25	-6.66555	23.67	1.794	0.5640
# A300 - A600	-6.00	4.74	25	-21.16555	9.17	-1.266	0.8606
# A300 - A1200	3.75	4.74	25	-11.41555	18.92	0.791	0.9837
# A600 - A1200	9.75	4.74	25	-5.41555	24.92	2.057	0.4064

```
#
# Confidence level used: 0.95
# Conf-level adjustment: tukey method for comparing a family of 7 estimates
# P value adjustment: tukey method for comparing a family of 7 estimates
```

Example 1 in R

Only one significant difference between population means can be found at 0.05 significance level, and that is between Control (μ_1) and A1200 (μ_7). The hypotheses for this test are below.

$$H_0 : \mu_1 - \mu_7 = 0$$

$$H_1 : \mu_1 - \mu_7 \neq 0.$$

The null hypothesis can be rejected with p-value of 0.006. The lower and upper bound allow any two-sided test of this contrast to be conducted; if the hypothesised difference is outside the CI then the null can be rejected. The important point is the family-wise error rate has been controlled - the chance of rejecting at least one hypothesis in error is no more than 0.05.

Example 1 in R

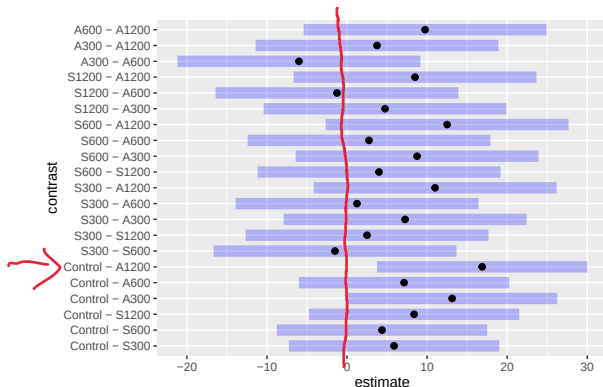
It is important to note that the Tukey CIs are wider than the individual CIs. For instance, the upper bound for an individual CI on the contrast A600-A1200 ($\mu_6 - \mu_7$) can be calculated with the code below.

```
# Individual 95% upper bound for mean of contrast A600-A1200.  
a <- 7 #numver of levels  
N <- 32 #overall sample size  
t975 <- qt(0.975, N-a) #0.975-quantile from T(N-a)  
9.75 + 4.74*t975 #95% upper bound  
# [1] 19.51221
```

Example 1 in R

We can also plot the CIs using the contrast object.

```
# Plot Tukey 95% CIs  
plot(fit.tukey)
```



We see that the only 95% CI that does not cover zero is the contrast Control-A1200 ($\mu_1 - \mu_7$).

Example 1 in R

Next, we use Fisher's Least Significance Difference (LSD) for comparison of all contrasts.

```
fit.fisher <- contrast(object=meansObj, method="pairwise", adjust="none")  
summary(fit.fisher, infer=c(T,T), level=0.95, side="two-sided")
```

# contrast	estimate	SE	df	lower.CL	upper.CL	t.ratio	p.value
# Control - S300	5.88	4.10	25	-2.5774	14.33	1.432	0.1647
# Control - S600	4.38	4.10	25	-4.0774	12.83	1.066	0.2966
# Control - S1200	8.38	4.10	25	-0.0774	16.83	2.041	0.0520
# Control - A300	13.12	4.10	25	4.6726	21.58	3.198	0.0037
# Control - A600	7.12	4.10	25	-1.3274	15.58	1.736	0.0949
# Control - A1200	16.88	4.10	25	8.4226	25.33	4.112	0.0004
# S300 - S600	-1.50	4.74	25	-11.2600	8.26	-0.317	0.7542
# S300 - S1200	2.50	4.74	25	-7.2600	12.26	0.528	0.6025
# S300 - A300	7.25	4.74	25	-2.5100	17.01	1.530	0.1386
# S300 - A600	1.25	4.74	25	-8.5100	11.01	0.264	0.7941
# S300 - A1200	11.00	4.74	25	1.2400	20.76	2.321	0.0287
# S600 - S1200	4.00	4.74	25	-5.7600	13.76	0.844	0.4066
# S600 - A300	8.75	4.74	25	-1.0100	18.51	1.846	0.0767
# S600 - A600	2.75	4.74	25	-7.0100	12.51	0.580	0.5669
# S600 - A1200	12.50	4.74	25	2.7400	22.26	2.638	0.0142
# S1200 - A300	4.75	4.74	25	-5.0100	14.51	1.002	0.3258
# S1200 - A600	-1.25	4.74	25	-11.0100	8.51	-0.264	0.7941
# S1200 - A1200	8.50	4.74	25	-1.2600	18.26	1.794	0.0850
# A300 - A600	-6.00	4.74	25	-15.7600	3.76	-1.266	0.2171
# A300 - A1200	3.75	4.74	25	-6.0100	13.51	0.791	0.4362
# A600 - A1200	9.75	4.74	25	-0.0100	19.51	2.057	0.0502
#							
# Confidence level used: 0.95							

Example 1 in R

Now, let us turn to Scheffe's method.

```
fit.scheffe <- contrast(object=meansObj, method="pairwise", adjust="scheffe")  
summary(fit.scheffe, infer=c(T,T), level=0.95, side="two-sided")
```

# contrast	estimate	SE	df	lower.CL	upper.CL	t.ratio	p.value
# Control - S300	5.88	4.10	25	-9.99	21.7	1.432	0.9081
# Control - S600	4.38	4.10	25	-11.49	20.2	1.066	0.9770
# Control - S1200	8.38	4.10	25	-7.49	24.2	2.041	0.6566
# Control - A300	13.12	4.10	25	-2.74	29.0	3.198	0.1614
# Control - A600	7.12	4.10	25	-8.74	23.0	1.736	0.8006
# Control - A1200	16.88	4.10	25	1.01	32.7	4.112	0.0316
# S300 - S600	-1.50	4.74	25	-19.82	16.8	-0.317	1.0000
# S300 - S1200	2.50	4.74	25	-15.82	20.8	0.528	0.9995
# S300 - A300	7.25	4.74	25	-11.07	25.6	1.530	0.8783
# S300 - A600	1.25	4.74	25	-17.07	19.6	0.264	1.0000
# S300 - A1200	11.00	4.74	25	-7.32	29.3	2.321	0.5117
# S600 - S1200	4.00	4.74	25	-14.32	22.3	0.844	0.9932
# S600 - A300	8.75	4.74	25	-9.57	27.1	1.846	0.7516
# S600 - A600	2.75	4.74	25	-15.57	21.1	0.580	0.9991
# S600 - A1200	12.50	4.74	25	-5.82	30.8	2.638	0.3587
# S1200 - A300	4.75	4.74	25	-13.57	23.1	1.002	0.9832
# S1200 - A600	-1.25	4.74	25	-19.57	17.1	-0.264	1.0000
# S1200 - A1200	8.50	4.74	25	-9.82	26.8	1.794	0.7756
# A300 - A600	-6.00	4.74	25	-24.32	12.3	-1.266	0.9471
# A300 - A1200	3.75	4.74	25	-14.57	22.1	0.791	0.9952
# A600 - A1200	9.75	4.74	25	-8.57	28.1	2.057	0.6481

```
#  
# Confidence level used: 0.95  
# Conf-level adjustment: scheffe method with rank 6  
# P value adjustment: scheffe method with rank 6
```

Example 1 in R

Note 1: The 95% upper bound for the contrast A600-A1200 ($\mu_6 - \mu_7$) of 19.51 using Fisher's method is the same as our manual calculation.

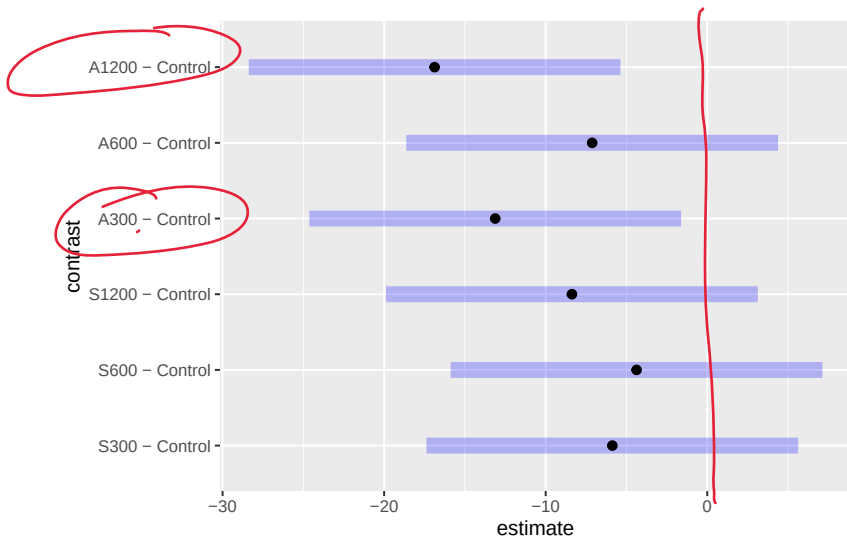
Note 2: The CIs from Scheffe's method, which are adjusted to control family-wise error rate, are wider than those produced with Tukey's method.

Next we examine Dunnett's method, which merely compares all treatment to a control group. By default, the first level of the factor is taken as the control group, in our case Control (this can be changed with the "ref" argument).

Example 1 in R

```
fit.dunnet <- contrast(object=meansObj, method="trt.vs.ctrl", adjust="mvt",  
  ref=1) #save contrast object (first level is default reference level)  
summary(fit.dunnet, infer=c(T,T), level=0.95, side="two-sided")  
# contrast      estimate  SE df lower.CL upper.CL t.ratio p.value  
# S300 - Control    -5.88 4.1 25   -17.4    5.63 -1.432  0.5852  
# S600 - Control    -4.38 4.1 25   -15.9    7.13 -1.066  0.8255  
# S1200 - Control   -8.38 4.1 25   -19.9    3.13 -2.041  0.2340  
# A300 - Control   -13.12 4.1 25   -24.6   -1.62 -3.198  0.0201  
# A600 - Control    -7.12 4.1 25   -18.6    4.38 -1.736  0.3874  
# A1200 - Control  -16.88 4.1 25   -28.4   -5.37 -4.112  0.0021  
#  
# Confidence level used: 0.95  
# Conf-level adjustment: mvt method for 6 estimates  
# P value adjustment: mvt method for 6 tests  
#  
plot(fit.dunnet) #plot 95% CIs
```

Example 1 in R



Example 1 in R

We see that Dunnett's method finds a significant difference in the contrast A1200-Control ($\mu_7 - \mu_1$), as did Tukey's method, but also between A300-Control ($\mu_5 - \mu_1$).

$$H_0 : \mu_5 - \mu_1 = 0$$

$$H_1 : \mu_5 - \mu_1 \neq 0.$$

The null hypothesis can be rejected with p-value 0.021. The CIs involving the Control contrast are wider in Tukey's method than in Dunnett's method basically because there are more CIs for which the family-wise error rate must be controlled.

Finally we consider comparison with a user-defined contrast list.

Example 1 in R

Custom contrasts using Scheffe's method.

contrlist <- list("Control v Treatment"=c(6,-1,-1,-1,-1,-1,-1),

"Spring v Autumn"=c(0,1,1,1,-1,-1,-1),

"Within Spring"=c(0,2,-1,-1,0,0,0),

"Within Autumn"=c(0,0,0,0,2,-1,-1)) #setup contrast list

fit.contrlist <- contrast(object=meansObj, method=contrlist,

adjust="scheffe") #save contrast object

summary(fit.contrlist, infer=c(T,T), level=0.95, side="two-sided") #view re

# contrast	estimate	SE	df	lower.CL	upper.CL	t.ratio	p.value
# Control v Treatment	55.75	16.42	25	1.22	110.3	3.396	<u>0.0431</u>
# Spring v Autumn	18.50	8.21	25	-8.77	45.8	2.254	0.3081
# Within Spring	1.00	8.21	25	-26.27	28.3	0.122	1.0000
# Within Autumn	-2.25	8.21	25	-29.52	25.0	-0.274	0.9993

Confidence level used: 0.95

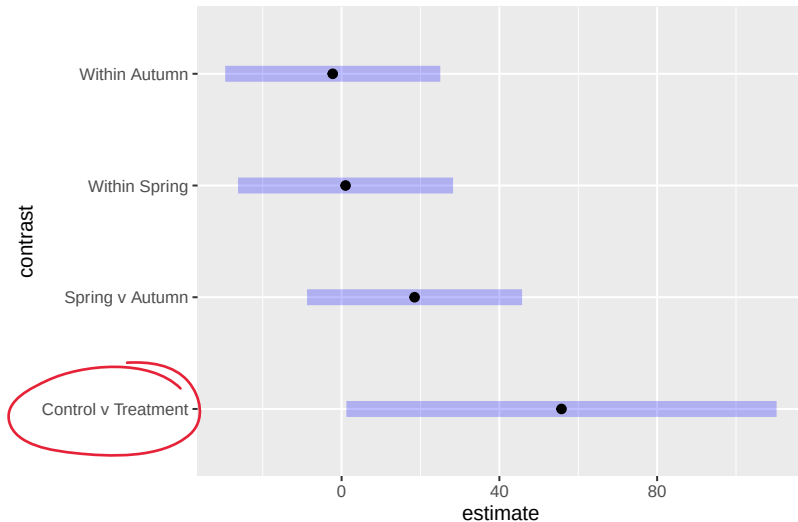
Conf-level adjustment: scheffe method with rank 4

P value adjustment: scheffe method with rank 4

plot(fit.contrlist) #plot 95% CIs

$$\begin{aligned} H_0: \mu_1 - \mu_2 - \mu_3 - \dots - \mu_7 &= 0 \\ \mu_1 &= \mu_2 + \mu_3 + \dots + \mu_7 \\ \mu_2 + \mu_3 + \mu_4 &= \mu_5 + \mu_6 + \mu_7 \end{aligned}$$

Example 1 in R



Example 1 in R

Only the Control v Treatment contrast is significantly different from zero.

$$H_0 : 6\mu_1 - \mu_2 - \mu_3 - \mu_4 - \mu_5 - \mu_6 - \mu_7 = 0$$

$$H_1 : 6\mu_1 - \mu_2 - \mu_3 - \mu_4 - \mu_5 - \mu_6 - \mu_7 \neq 0.$$

The null hypothesis can be rejected with p-value of 0.0431.

Example 2

The data in the table below are the melting times in seconds for three different brands of margarine (coded 1—3) and one brand of butter (coded 4). The butter was used for comparison purposes. The sizes and shapes of the initial margarine/butter pats were as similar as possible, and these were melted one by one in a clean frying pan over a constant heat.

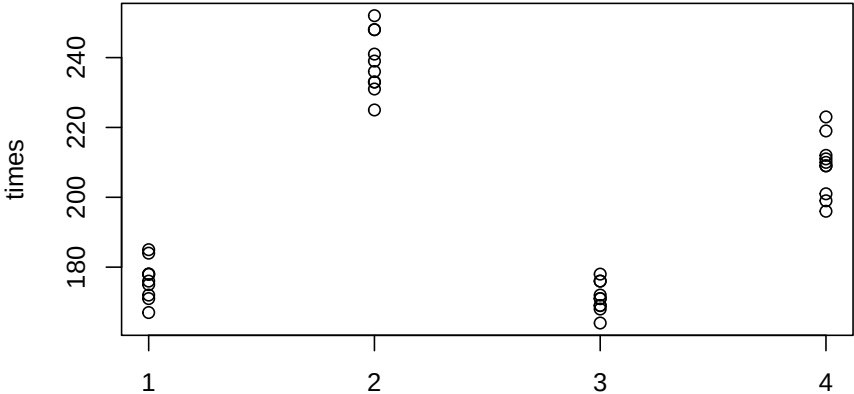
Margarine 1		Margarine 2		Margarine 3		Butter	
167	176	231	225	176	176	201	223
171	185	233	241	168	169	199	209
178	172	236	248	171	164	196	219
175	178	252	239	172	169	211	212
184	178	233	248	178	171	209	210

Example 2 in R

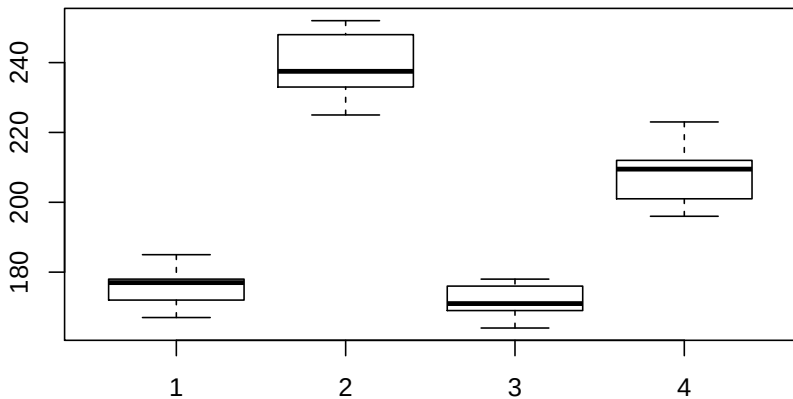
We start with loading data and producing a box-plot.

```
times <- c(167,176,231,225,176,176,201,223,  
          171,185,233,241,168,169,199,209,  
          178,172,236,248,171,164,196,219,  
          175,178,252,239,172,169,211,212,  
          184,178,233,248,178,171,209,210)  
brand <- rep(rep(c("1","2", "3","4"), each=2), 5)  
melt <- data.frame(brand, times)  
#  
# Plot data  
stripchart(times~brand, vertical=TRUE, pch=1, data=potato)  
boxplot(times~brand, data = melt)
```


Example 2 in R



Example 2 in R



Example 2 in R

$$H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4$$

Run ANOVA and perform diagnostics.

```
# Fit ANOVA model
fit <- aov(times~brand, data=melt)
summary(fit)
```

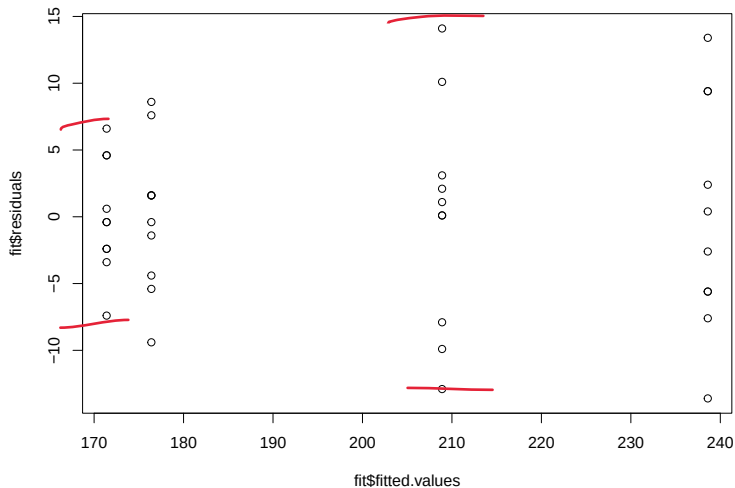
#	Df	Sum Sq	Mean Sq	F value	Pr(>F)
# brand	3	29386	9795	200.3	<2e-16 ***
# Residuals	36	1760	49		
# ---					

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

reject H₀.

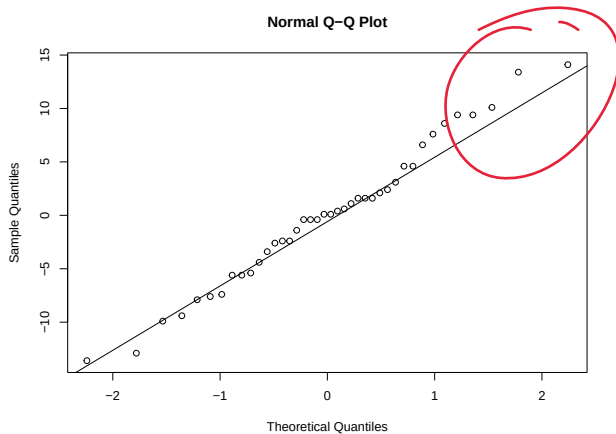
Example 2 in R

```
plot(fit$residuals~fit$fitted.values)
```



Example 2 in R

```
qqnorm(fit$residuals)  
qqline(fit$residuals)
```



Example 2 in R

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$$

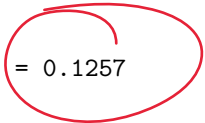

$$H_0 : \tau_1 = \tau_2 = \tau_3 = \tau_4 = 0$$

$$H_1 : \text{at least one } \tau_i \neq 0.$$

The null hypothesis can be rejected with extremely small p-value and we conclude at least one treatment effect is non-zero (or equivalently at least one mean is different from the others). Again it looks like there is a problem with increasing variance.

Example 2 in R

```
# Testing equal variances
library(lawstat) # install.packages("lawstat") #install if necessary
levene.test(melt$times, melt$brand,
            location="median", correction.method="zero.correction")
# Modified robust Brown-Forsythe Levene-type test based on the absolute
# deviations from the median with modified structural zero removal
method
# and correction factor
#
# data: melt$times
# Test Statistic = 2.0562, p-value = 0.1257
```



Looks OK.

Example 2 in R

What about normality?

```
res <- residuals(fit)
# Shapiro-Wilk test.
shapiro.test(res)
# Shapiro-Wilk normality test
#
# data:  res
# W = 0.98334, p-value = 0.8104
#
# Cramer-von Mises test.
library(nortest) # install.packages("nortest") #install if necessary
nortest::cvm.test(res)
# Cramer-von Mises test.
#
# data:  res
# W = 0.040671, p-value = 0.6601
```


Example 2 in R

```
# Anderson-Darling test.  
nortest::ad.test(res)  
# data:  res  
# A = 0.23773, p-value = 0.7688
```

All of the above tests indicate there is no problem with the normality assumption.

Example 2 in R

Now define contrasts and compare them.

```
# Custom contrasts using Scheffe's method
library("emmeans") #install.packages("emmeans") #install if necessary
contrlist <- list("Butter v Margarine"=c(1,1,1,-3),
  "Within Margarine 1"=c(-1,0,1,0),
  "Within Margarine 2"=c(-1,2,-1,0))
meansObj <- emmeans(fit, "brand")
fit.contrlist <- contrast(object=meansObj, method=contrlist,
  adjust="scheffe")
summary(fit.contrlist, infer=c(T,T), level=0.95, side="two-sided")
```

# contrast	estimate	SE	df	lower.CL	upper.CL	t.ratio	p.value
# Butter v Margarine	-40.3	7.66	36	-62.8	-17.84	-5.261	<u>0.0001</u>
# Within Margarine 1	-5.0	3.13	36	-14.2	4.17	-1.599	0.4746
# Within Margarine 2	129.4	5.42	36	113.5	145.28	23.891	<u><.0001</u>

```
#
# Confidence level used: 0.95
# Conf-level adjustment: scheffe method with rank 3
# P value adjustment: scheffe method with rank 3
```

Example 2 in R

There are two significant differences: Butter v Margarine and Within Margarine 2.
So

$$(1, 1, 1, -3)$$

$$H_0 : \mu_1 + \mu_2 + \mu_3 - 3\mu_4 = 0$$

$$\frac{\mu_1 + \mu_2 + \mu_3}{3} = \mu_4.$$

and

$$H_0 : -\mu_1 + 2\mu_2 - \mu_3 = 0$$

can both be rejected.

$$2\mu_2 = \mu_1 + \mu_3$$
$$\mu_2 = \frac{\mu_1 + \mu_3}{2}$$

Example 2 in R

$$\binom{4}{2} = 6.$$

Finally apply Tukey's method.

```
fit.tukey <- contrast(object=meansObj, method="pairwise",  
  adjust="tukey")  
summary(fit.tukey, infer=c(T,T), level=0.95, side="two-sided")
```

#	contrast	estimate	SE	df	lower.CL	upper.CL	t.ratio	p.value
#	1 - 2	-62.2	3.13	36	-70.62	-53.8	-19.891	<.0001
#	1 - 3	5.0	3.13	36	-3.42	13.4	1.599	0.3920
#	1 - 4	-32.5	3.13	36	-40.92	-24.1	-10.393	<.0001
#	2 - 3	67.2	3.13	36	58.78	75.6	21.490	<.0001
#	2 - 4	29.7	3.13	36	21.28	38.1	9.498	<.0001
#	3 - 4	-37.5	3.13	36	-45.92	-29.1	-11.992	<.0001
#								
#	Confidence level used: 0.95							
#	Conf-level adjustment: tukey method for comparing a family of 4 estimates							
#	P value adjustment: tukey method for comparing a family of 4 estimates							

All mean combinations other than μ_1 and μ_3 are significantly different.